

PLP (Power Learn Project)

Python Module

Final project

Submitted by

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I. Title & Introduction

COVID-19 Vaccination Analysis

This notebook analyzes global COVID-19 vaccination progress, showing rollout speed, vaccination rates, and geographic trends.

II. Key Insights

1. **India leads in absolute vaccination numbers**, administering hundreds of millions of doses,
2. yet its large population means full vaccination coverage is still below 70%.
3. **Smaller nations like Seychelles and Malta vaccinated their populations quickly**,

achieving >70% coverage earlier than most other countries.

4. **High-income countries lead global vaccination rollout**,

whereas many low-income countries remain below 20% coverage, highlighting a stark inequality.

5. **Vaccine rollout acceleration is visible in late 2021 and early 2022**,

coinciding with global supply chain improvements and booster campaigns.

6. **Data gaps and missing values** remain a concern; several countries do not report consistently,

leading to incomplete coverage comparisons.

III. Highlight Anomalies or Patterns

- **Seychelles** stands out as an early vaccination leader,
- reaching over 60% full vaccination coverage before larger nations.
- Some countries, like **Tanzania and North Korea**,

have very sparse or no vaccination reporting, creating blind spots in global tracking.

- Nations like **Israel and the United Arab Emirates** were early adopters, rapidly vaccinating large portions of their populations by mid-2021.

- Despite having the highest total doses administered,

India and the United States show slower population percentage growth

due to their large populations.

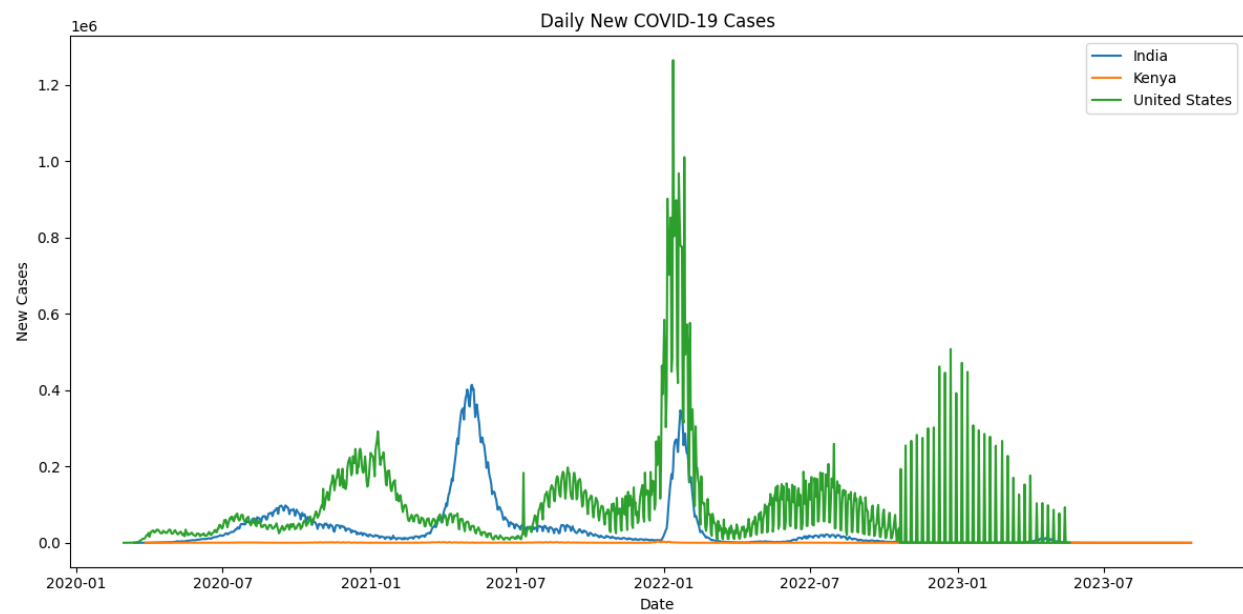
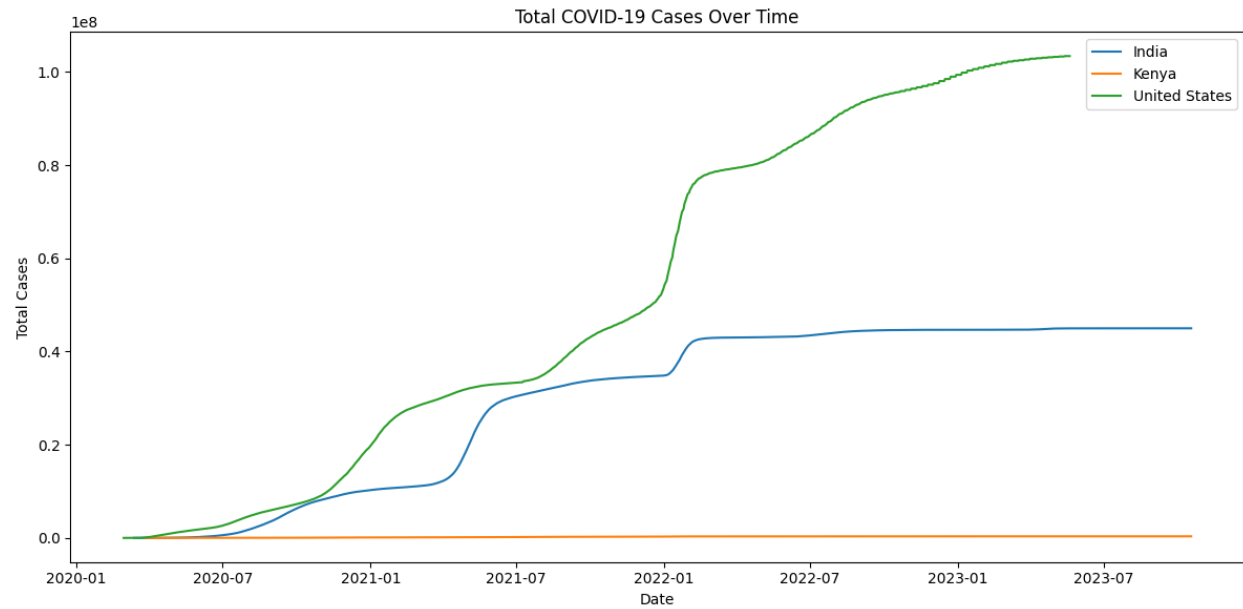
- Several **low-income African nations** report vaccination rates below 10%, underscoring inequitable vaccine access globally.

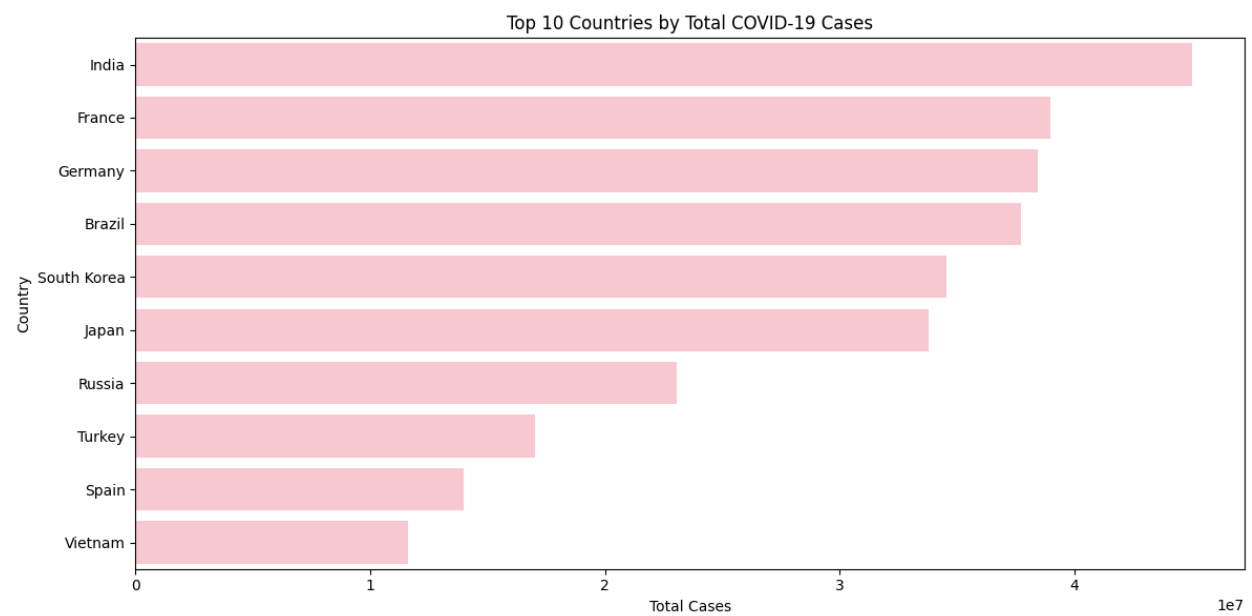
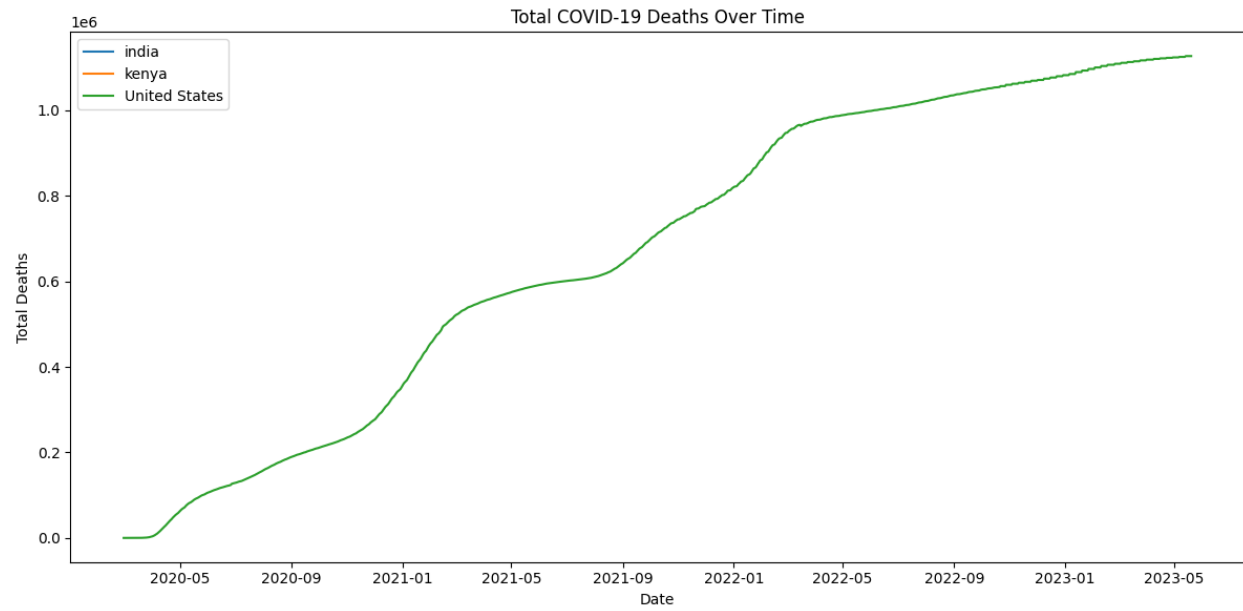
IV. Visualizations Section

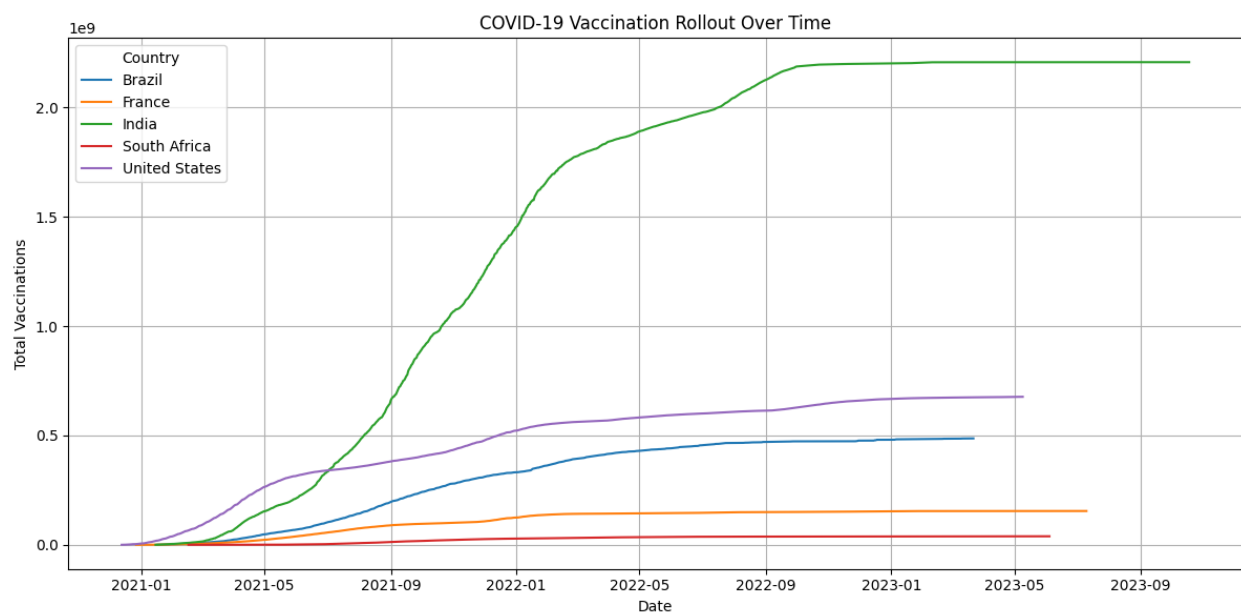
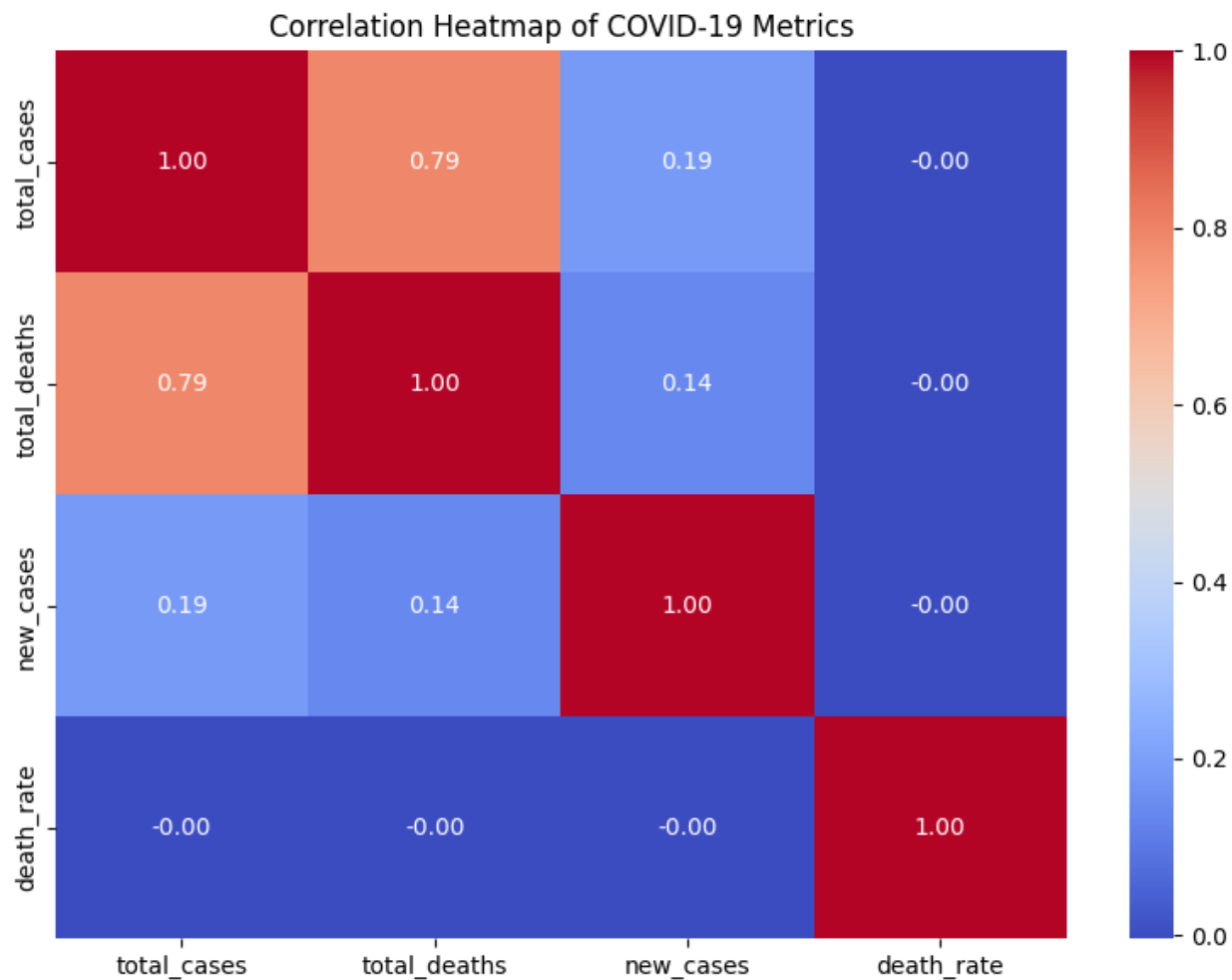
Below are the visualizations illustrating vaccination trends:

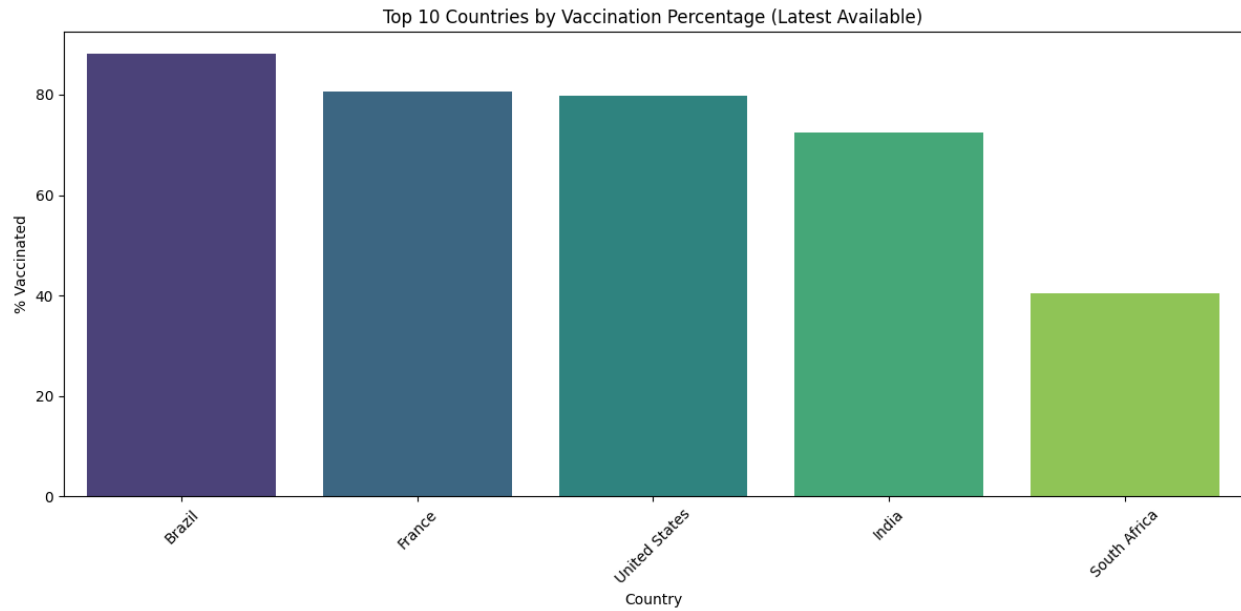
1. **Line Chart** – Shows cumulative vaccinations over time for selected countries
2. **Bar Chart** – Displays top 10 countries by vaccination percentage
3. **Choropleth Map** – Global vaccination coverage (latest available date)

These visuals make disparities in vaccine access and rollout speed clearer.

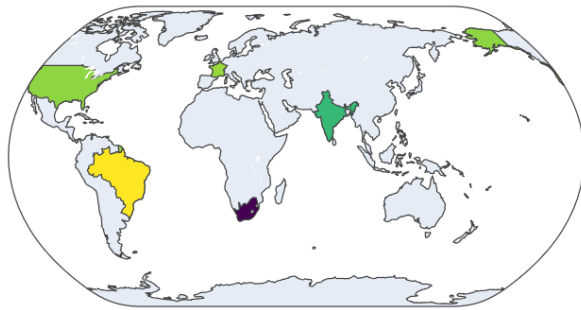








Global COVID-19 Vaccination Rates (Latest Available)



V. Conclusion

The data highlights clear inequalities in global vaccination distribution.

Smaller nations with strong supply chains achieved rapid coverage,

while low-income countries lag far behind.

Key takeaways:

- Wealthier nations drove early vaccine adoption
- Smaller countries often reached full coverage faster

- Data gaps remain a challenge for accurate global comparison

This analysis can guide further research into vaccine equity, supply chain logistics, and policy interventions to ensure better preparedness for future health crises.